

BUS33-1

Advanced Marketing

Albert School — 22 hours

Preface

In Marketing Fundamentals (BUS13-1) you learned that marketing is a discipline of decisions made under scarcity: choosing which demand to pursue (Where to Play), why you deserve to be chosen (How to Win), and where to concentrate resources (How to Allocate). You learned to segment by the job a customer is trying to do, to write a position that **constrains**, and to price to perceived **value**. In Sales, Negotiation & Client Relations (BUS23-1) you carried that strategy into the conversation — the pipeline, the objection, the price across the table. And in SQL & Data Querying (DAT22-5) you learned to turn a business question into a query, run it, and read the result.

This course is where those three threads braid together. The S1 toolkit told you **what** to do; it was largely silent on **how much it costs to acquire a customer, whether a tactic actually worked**, and **what a customer is worth over time**. Advanced Marketing adds those two missing layers: a **data layer** — CAC/LTV economics, attribution, cohorts, experimentation — and a **cultural/prospective layer** — why people consume, and how to tell a durable structural trend from a fad. Together they turn marketing from a menu of tactics into **allocation decisions defended with evidence**.

The shift is concrete. A junior marketer says “let’s do influencer marketing, it’s working for everyone.” A marketer trained here says “our blended CAC on that channel would be about €70, our 12-month contribution per customer is €110, the channel saturates at roughly 4,000 customers a quarter, and here is the A/B test that would tell us within three weeks whether the early read is real.” Same energy; entirely different standard of proof.

We assume the three prerequisites throughout. Segmentation, positioning, the 4Ps, BATNA and the sales funnel, **GROUP BY** and joins — these are settled tools we **use**, not topics we re-teach. What is new is the measured, compounding, evidence-bound craft of **growth**. A warning to carry from the start: almost every number in this book can be gamed, and almost every framework can be cargo-culted. The discipline is not knowing the acronyms — anyone can learn AARRR — but knowing which number is lying to you and why.

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1 The AARRR Growth Framework

1.1 One company, five different problems

A meal-kit startup has 50,000 people visit its site each month. Of those, 5,000 create an account, 1,500 complete a first order, 600 are still ordering three months later, and 50 refer a friend. Revenue is flat and the founder’s instinct is “we need more traffic.” But look again: the site already attracts 50,000 people and converts only 3% of them to a first order, and keeps only 40% of those to month three. Pouring money into the top of this system fills a leaking bucket. The useful question is not “how do we get more visitors?” but “**which of the five stages is actually losing us the most growth?**” AARRR is the framework that forces that question.

1.2 The five stages

Definition 1 (AARRR — the pirate metrics) AARRR decomposes a customer’s journey into five measurable stages, each with its own conversion metric:

1. **Acquisition** — the user arrives (visits, signups, app installs). Metric: volume and cost per arrival by channel.
2. **Activation** — the user has a first valuable experience (the “aha moment”: first order placed, first project created). Metric: % of acquired users who activate.
3. **Retention** — the user comes back and keeps using the product. Metric: % still active after N days/weeks; this is the single best predictor of a real business.
4. **Revenue** — the user generates money (purchase, subscription, upgrade). Metric: conversion to paying, average revenue per user.
5. **Referral** — the user brings in other users (invites, shares, word of mouth). Metric: referrals sent and the fraction that convert.

Each stage is a filter on the one before it, so growth is the **product** of the five conversion rates — improving the weakest stage usually beats adding volume at the top.

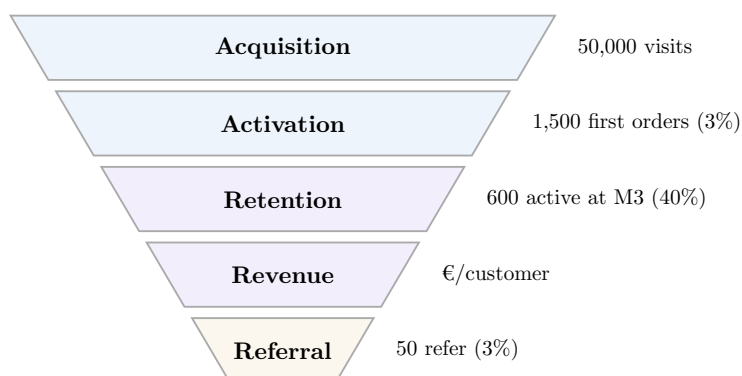


Figure 1: The AARRR funnel for the meal-kit startup. Width is roughly proportional to the surviving population. The steepest drop is Acquisition $\rightarrow$ Activation (only 3% place a first order), so that stage — not more top-of-funnel traffic — is where a euro of effort buys the most growth.

Worked example — Diagnosing where to spend the next euro. The founder above is choosing between (a) €20k of extra ads to lift visits from 50,000 to 70,000, and (b) €20k of UX work to lift Acquisition $\rightarrow$ Activation from 3% to 5%. Option (a) at the current 3% yields $70,000 \times 0.03 = 2,100$ first orders, up from 1,500 — about 600 more. Option (b) at the current 50,000 visits yields $50,000 \times 0.05 = 2,500$ first orders — 1,000 more, for the same spend, and the gain **compounds** because every downstream stage now starts from a larger

base. The framework converts a vague “we need growth” into a ranked, quantified allocation decision.

Common pitfall AARRR is a **diagnostic**, not a to-do list. The beginner reads the five letters and tries to “do all of them” at once. The discipline is the opposite: measure all five, then concentrate effort on the **one** stage whose improvement most raises the end-to-end product. A startup with great acquisition and terrible retention does not have an acquisition problem to solve — adding users to a leaky bucket just raises the cost of the leak.

1.3 Retention is the load-bearing stage

Of the five, retention deserves special weight, because it silently sets the ceiling on every other stage. A product people abandon cannot be saved by acquisition (you refill a bucket faster than it drains, at rising cost), cannot sustain revenue (no repeat purchases), and generates no referral (people do not recommend what they quit). This is why experienced growth operators say “retention is the only metric that matters” — an overstatement, but a corrective to the beginner’s obsession with acquisition.

Remark Notice how AARRR operationalises the “How to Allocate” decision from BUS13-1. There the allocation was across the 4Ps; here it is across five conversion stages. Same logic — finite resources, ranked by marginal return — applied to the customer lifecycle. We will compute the marginal returns precisely once we have CAC and LTV (Chapters 2 and 6).

Exercises

1. A mobile game has 200,000 installs/month, 30% reach the tutorial-complete “activation” event, 20% of those are still playing at day 7, 8% of day-7 players ever pay, and each payer invites on average 0.4 friends who install. Compute the number of activated, retained, and paying users, and identify the single stage where a 50% relative improvement adds the most paying users.
2. For a product you know, define a concrete, measurable “activation” event — the first moment a user gets real value — and justify why that event, and not signup, is the right activation metric.
3. Explain, in two or three sentences, why a company with excellent acquisition and poor retention should usually pause acquisition spend before fixing retention.

Chapter takeaways. AARRR decomposes growth into Acquisition, Activation, Retention, Revenue, Referral — five conversion stages whose **product** is overall growth. Diagnose all five, then concentrate effort on the weakest, because gains there compound through every later stage. Retention is load-bearing: it caps what acquisition, revenue, and referral can ever achieve. AARRR is the “How to Allocate” discipline of BUS13-1 applied to the customer lifecycle.

2 Acquisition-Channel Economics

2.1 Why “what’s working for everyone” is the wrong question

A DTC skincare brand and a B2B payroll-software company both hear that “TikTok is where growth is.” For the skincare brand — €30 product, impulse purchase, visual, mass consumer audience — it may be superb. For the payroll company — €40k annual contract, six-person buying committee, bought after careful evaluation — it is almost certainly a waste: the audience does not buy payroll software off a 15-second video. Channels are not universally good or bad; each has **economics** and **viability conditions** that depend on the price point, margin, sales cycle, and audience of the specific business. This chapter makes those conditions explicit.

2.2 CAC and the unit-economics constraint

Definition 2 (Customer Acquisition Cost (CAC)) CAC is the total sales-and-marketing cost of acquiring one new customer over a period, divided by the number of new customers acquired in that period:

$$\text{CAC} = \frac{\text{total acquisition spend}}{\text{new customers acquired}}.$$

Blended CAC divides total spend by **all** new customers (including organic); **paid CAC** counts only paid spend and the customers it brought. Blended CAC flatters channels by crediting them with free organic customers, so channel decisions must use paid (marginal) CAC.

The number CAC must be judged against is the value a customer brings — their lifetime value, LTV (computed properly in Chapter 6). The central constraint of all paid acquisition is simply stated:

Definition 3 (The LTV/CAC viability condition) A paid acquisition channel is viable only if the value of a customer exceeds the cost of acquiring one — i.e. $\text{LTV} > \text{CAC}$. The widely used rule of thumb is

$$\frac{\text{LTV}}{\text{CAC}} \geq 3,$$

with a **CAC payback period** (months of contribution margin needed to repay CAC) short enough for the company’s cash position — often under 12 months. A ratio near 1 means you buy customers at cost (no room for overhead or profit); a ratio far above 3 often means you are **under-investing** and leaving growth unbought.

Common pitfall Two CAC mistakes recur. First, comparing CAC to **revenue** instead of **contribution margin**: a €100 sale at 20% margin contributes €20, so a €40 CAC loses money even though CAC is “less than revenue.” Always compare CAC to margin-based LTV. Second, using **blended CAC** to justify a **paid** channel: if organic already brings 1,000 free customers, dumping them into the denominator makes any paid channel look cheap. Decisions about whether to scale a paid channel need that channel’s **marginal CAC**.

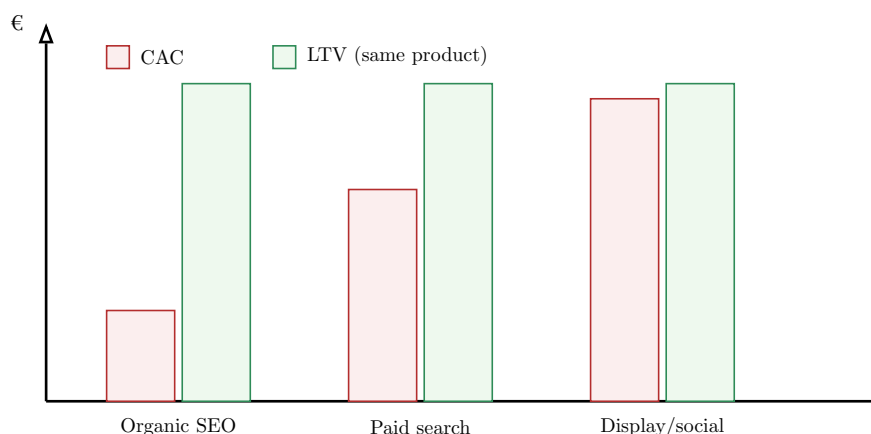


Figure 2: The same product (fixed LTV, green) acquired through three channels with different CAC (red). Organic SEO clears the $LTV/CAC \geq 3$ bar comfortably; paid search is marginal; display/social barely beats break-even. Viability is a property of the **channel–business pair**, not the channel alone.

2.3 The main channels and their viability conditions

Definition 4 (Acquisition channel families)

- **Paid search** (Google Ads): **high intent** — the user is already searching for a solution. CAC scales predictably but rises as you bid on broader terms. Viable when customers actively search for your category and LTV supports the per-click cost.
- **Paid social** (Meta, TikTok): **interruption** — you reach users not actively looking. Excellent for visual, impulse, mass-consumer products; weak for high-consideration B2B. CAC depends heavily on creative quality.
- **Organic / SEO & content**: low marginal CAC but **slow** and **compounding** — content keeps attracting traffic for years. Viable when the category is searched and you can sustain the upfront investment before payoff.
- **Partnerships / channel**: acquire through another company’s audience (integrations, resellers, affiliates). Viable when a partner already holds your customers’ trust and incentives align.

Each family has a different **CAC curve**, **speed**, and **audience fit**; the choice is a portfolio decision, not a single bet.

Worked example — Matching channels to two businesses. **DTC skincare (€30, impulse, visual)**: paid social leads — strong creative on Meta/TikTok reaches the mass audience cheaply; paid search captures the few who already search “vitamin C serum”; SEO/content compounds over time. B2B partnerships are irrelevant. **B2B payroll software (€40k/yr, considered)**: paid search on high-intent terms (“payroll software for 50 employees”) and partnerships with accounting firms dominate; paid social is mostly wasted because the buyer does not impulse-purchase a €40k system. Same menu of channels, opposite ranking — driven by price, intent, and audience (Learning Outcome 1).

Common pitfall Beware the **channel-fashion trap**: choosing a channel because it is trending rather than because its economics fit your business. The right test is always the LTV/CAC condition **for your unit economics and audience**, computed on this channel’s

marginal CAC — not the fact that a competitor or an influencer says it works. A channel that is wildly profitable for a €30 impulse product can be ruinous for a €40k considered purchase, and vice versa.

2.4 Channels saturate

A final economic fact that breaks naïve plans: CAC is not constant. The first customers from a channel are the cheapest (the most eager, the best-targeted); as you scale spend, you reach less-interested people and CAC **rises**. Every channel has a volume beyond which the marginal CAC exceeds LTV — its **saturation point**. This is why no single channel scales forever and why growth plans (Chapter 9) allocate across a **portfolio**.

Exercises

4. A subscription box has a €45/month price, 60% contribution margin, and average customer lifespan of 10 months. Compute contribution-margin LTV. Then decide which of two channels is viable: Channel A at €120 CAC, Channel B at €230 CAC. Use the $LTV/CAC \geq 3$ rule and state the payback period for each.
5. Explain why blended CAC can make a failing paid channel look profitable, and describe exactly which number you would compute instead to make a scaling decision.
6. Pick a real product and rank paid search, paid social, SEO/content, and partnerships for it. Justify the ranking using price point, purchase intent, sales-cycle length, and audience — not channel popularity.

Chapter takeaways. CAC is acquisition spend per new customer; use **paid, marginal CAC** for channel decisions, not blended CAC, and compare it to **contribution-margin LTV**. A channel is viable when $LTV/CAC \geq 3$ with an acceptable payback period. Paid search, paid social, organic/SEO, and partnerships have different CAC curves, speeds, and audience fits, so the right channel is a property of the channel–business pair. Channels saturate — CAC rises with volume — so acquisition is a portfolio, not a single bet.

3 Growth Loops vs Funnels

3.1 Two ways to grow, only one of which compounds

The AARRR funnel (Chapter 1) is a **linear** picture: pour users in the top, some fraction reach the bottom. To keep growing, you must keep pouring — and the pouring (acquisition) costs money that rises as channels saturate. A **growth loop** is a different shape: the **output** of the system feeds back as new **input**, so growth produces more growth without proportional new spend. Understanding when you have a funnel and when you have a loop is the difference between a business that must buy every customer and one that partly grows itself.

Definition 5 (Funnel vs growth loop) A **funnel** is a one-directional sequence of conversion stages: inputs enter at the top, a fraction exits as customers at the bottom, and sustaining growth requires continuously feeding the top. A **growth loop** is a closed cycle in which the output of one cycle becomes the input of the next — e.g. **new users** → **they create content / invite others** → **that attracts more new users**. A funnel is a

leaky bucket (you must keep refilling); a loop is a **compounding mechanism** (each turn enlarges the next).

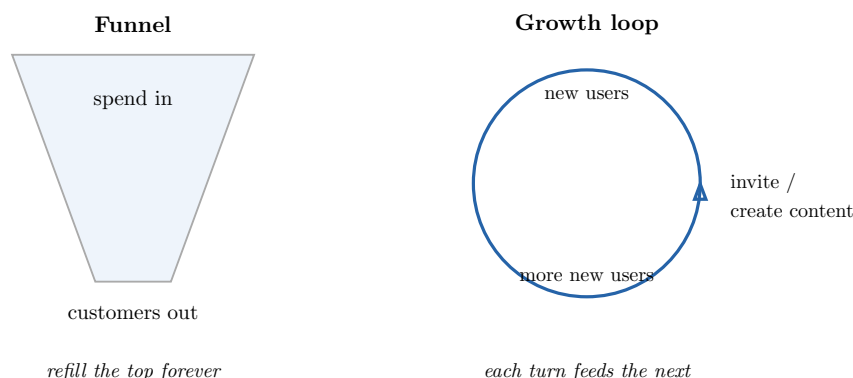


Figure 3: A funnel must be refilled from outside every cycle; a growth loop reinvests its output as next cycle's input. Loops compound; funnels do not.

3.2 A worked contrast

Worked example — Same product, funnel vs loop. Funnel version: a language-learning app buys installs on Meta. 10,000 installs → 2,000 activate → 200 subscribe. To get 200 more subscribers next month, it must buy 10,000 more installs. Growth is bought, linearly, forever.

Loop version: the same app adds a feature where learners share their progress and challenge friends. Now 2,000 activated users invite on average 1.3 friends, some of whom install, activate, and invite **their** friends. If each user brings more than one new activated user, the user base grows on its own; if each brings a fraction, the loop **amplifies** paid acquisition (every bought user drags in extra free ones), lowering blended CAC. The mechanism — not the spend — is doing the work.

Definition 6 (The viral coefficient (k)) For a referral loop, the **viral coefficient** k is the average number of new activated users each existing user generates: $k = i \times c$, where i is invitations sent per user and c is the conversion rate of an invitation. If $k \geq 1$, the loop is self-sustaining (each user more than replaces themselves) and growth is explosive; if $k < 1$, the loop is **sub-critical** but still valuable, because it multiplies every acquired user by a factor of $\frac{1}{1-k}$.

Worked example — Amplification below $k = 1$. Suppose paid channels bring 1,000 activated users a month and the referral loop has $k = 0.5$. Those 1,000 bring 500, who bring 250, who bring 125... The total is $\frac{1000}{1-0.5} = 2\{, \}000$ activated users — the loop **doubled** paid acquisition without doubling spend. A loop need not be viral ($k \geq 1$) to be economically decisive; even $k = 0.5$ halves effective CAC.

3.3 Loops have constraints that break them at scale

The seductive error is to assume a loop, once started, grows forever. It never does. Every loop runs into a **constraint** — a finite resource it depends on — that flattens it at scale.

Definition 7 (The loop constraint) A **growth-loop constraint** is the finite resource or saturating condition that causes the loop’s effective coefficient to fall as it scales — for example: the **addressable network** runs out (everyone’s friends already use the app); invitation **fatigue** (users stop inviting, or recipients stop accepting); **channel saturation** of the medium the loop rides on; or a **quality** ceiling (more content but lower average quality, so c falls). Identifying this constraint is what separates a defensible growth model from wishful exponential projection.

Common pitfall Two opposite errors. (1) **Mistaking a funnel for a loop**: calling any referral scheme a “growth loop” when its k is so low and its constraint so tight that it is really just a funnel with a trickle of word-of-mouth. (2) **Assuming a loop compounds forever**: projecting early exponential growth straight off the chart without naming the constraint (network exhaustion, fatigue, saturation) that will bend the curve down. A credible growth model states both the loop **and** the constraint that will eventually break it.

Remark Funnels and loops are not rivals — most real businesses are a funnel **with** one or more loops layered on. The funnel describes how a single cohort converts (Chapter 1); the loop describes how cohorts feed the next. Chapter 9’s growth plan must specify both: the funnel conversion rates **and** the loops that amplify them, each with its constraint.

Exercises

7. A note-taking app has users who invite on average 0.8 colleagues, of whom 40% activate. Compute k . Is the loop self-sustaining? By what factor does it amplify paid acquisition? If paid brings 5,000 activated users a month, what is the total with the loop?
8. For a marketplace (e.g. a ride-hailing or freelance platform), describe a growth loop that does **not** rely on invitations (hint: think about how supply and demand attract each other). Name the constraint that will slow it at scale.
9. Take a company you believe has a “growth loop” and argue either that it is a genuine compounding loop or merely a funnel with marketing attached. Identify the single constraint most likely to break or cap the loop.

Chapter takeaways. A funnel is a leaky bucket you must refill from outside; a growth loop reinvests its output as next-cycle input and therefore compounds. The viral coefficient $k = i \times c$ measures a referral loop: $k \geq 1$ is self-sustaining, and even $k < 1$ amplifies paid acquisition by $\frac{1}{1-k}$. Every loop has a **constraint** — network exhaustion, fatigue, saturation, quality ceiling — that breaks it at scale, and a credible model names it. Real businesses combine funnels and loops; the growth plan must specify both.

4 A/B Testing and Experimentation Discipline

4.1 Why “it went up after we changed it” proves nothing

A team changes the checkout button from grey to green and conversion rises from 4.0% to 4.3% the next week. Did the colour cause it? Maybe — or maybe that week had a payday, a viral post, better weather, or simply **noise**: flip a fair coin 1,000 times twice and the two runs will not

match. Without a **controlled comparison** and a **pre-committed rule** for reading it, every change can be declared a success after the fact, and the team learns nothing. The A/B test is the instrument that turns “it went up” into “the change caused a lift we can trust.”

Definition 8 (A/B test (controlled experiment)) An **A/B test** randomly splits comparable users into a **control** group (A, current experience) and a **treatment** group (B, the change), then compares a pre-chosen metric between them. Random assignment makes the groups statistically equivalent on everything **except** the change, so a difference in the metric can be attributed to the change rather than to confounders.

4.2 Statistical significance vs effect size

Definition 9 (Significance and effect size) **Statistical significance** (often a p -value below a threshold such as 0.05) asks: **is the observed difference unlikely to be mere chance?** **Effect size** asks a different question: **how big is the difference, and does it matter for the business?** The two are independent — a difference can be highly significant yet trivially small, or large yet not significant (too little data). A decision needs **both**: confidence that the effect is real **and** that it is big enough to act on.

Common pitfall The most common misreading in experimentation is treating a small p -value as proof of an important effect. With a large enough sample, a 0.05% conversion improvement can be “highly significant” ($p < 0.001$) and yet not worth the engineering cost. Always report the **effect size and its confidence interval**, not just whether $p < 0.05$. Significance tells you the effect is **real**; only effect size tells you it is **worth it**.

4.3 Statistical power and pre-committed sample size

Definition 10 (Statistical power and minimum detectable effect) **Statistical power** is the probability that a test will detect a real effect of a given size when one exists (conventionally targeted at 80%). It rises with sample size and with the **minimum detectable effect** (MDE) you care about. Before running a test you must compute the **required sample size** from three inputs: the baseline rate, the MDE you want to be able to detect, and the desired power and significance level. Running an **underpowered** test — too few users to detect the effect you care about — wastes time and produces a false “no difference.”

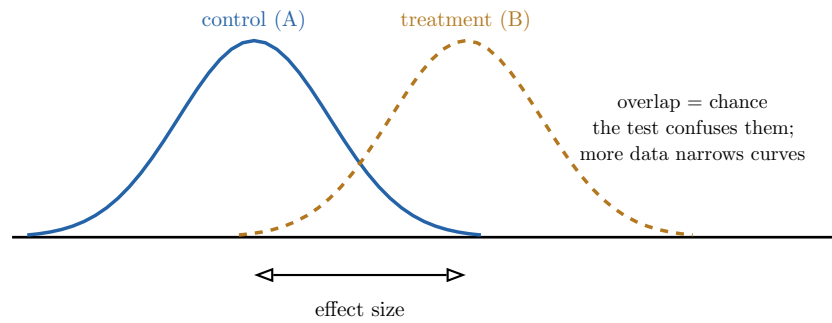


Figure 4: Each group’s measured rate is itself uncertain (a distribution, not a point). Power is the chance the two distributions separate enough to call the difference real. A bigger true effect (more separation) or more data (narrower curves) both raise power. Fix the required sample size **before** running.

4.4 Pre-committed decision rules and the peeking trap

Definition 11 (Pre-committed decision rule) A **pre-committed decision rule** fixes, **before** data collection begins, exactly what will be measured, on how many users, for how long, and what result will count as a win, a loss, or no-decision. Committing in advance prevents the experimenter from rationalising whatever the data happen to show — the discipline that makes the read defensible.

Common pitfall Peeking (or “*p*-hacking”) is the cardinal sin of experimentation: repeatedly checking results and stopping the moment they cross $p < 0.05$. Because the metric fluctuates, it will **eventually** cross the line by chance even when there is no real effect — stopping on that moment manufactures false positives. Equally corrupt: running the test, getting a null, then slicing by country / device / time until **some** subgroup is “significant.” Fix the sample size and the metric in advance, and read the test once, at the end.

Worked example — A disciplined A/B test, end to end. An e-commerce team wants to test a one-page checkout. **Hypothesis:** it lifts checkout-completion from a baseline 4.0%. **MDE:** they care only about a lift of at least 0.5 percentage points (to 4.5%). **Pre-committed:** a power calculation at 80% power, $\alpha = 0.05$ says they need 31,000 users per arm; they will run until both arms reach that, about three weeks, and read **once**. **Falsifier:** “if treatment completion is not at least 0.4 pts above control with the interval excluding zero, we ship nothing.” Result: 4.0% vs 4.6%, interval [0.3, 0.9] pts, $p = 0.004$. The effect is significant **and** above the MDE, so they ship — and they can say exactly what evidence **would** have killed it.

4.5 Diagnosing funnel drop-off with experiments

Experiments are also the tool for fixing the AARRR funnel (Chapter 1). When a stage leaks — say Activation — you locate the worst-converting step, form a hypothesis about **why** users drop, and test a change against control. The funnel tells you **where** to intervene; the A/B test tells you **whether** the intervention actually worked, rather than coincided with an unrelated swing.

Remark This is the experimentation cousin of the sales-funnel diagnosis in BUS23-1: there you found the two highest-leverage stage transitions; here you fix them with controlled tests

instead of intuition. “Find the leak” (diagnosis) and “prove the fix” (experiment) are two halves of the same allocation discipline.

Exercises

10. A landing page converts at 5%. A team runs a test, peeks daily, and stops on day 4 when p first dips below 0.05 with the variant at 5.4%. Explain precisely why this result is untrustworthy and what they should have done instead.

11. Distinguish significance from effect size using a concrete example where a result is statistically significant but not worth shipping, and one where a promising effect is not yet significant. Say what you would do in each case.

12. Design a complete A/B test for one change to a product you know: state the hypothesis, the metric, the MDE, roughly why the sample must be large, the pre-committed decision rule, and — crucially — what observed result would have **falsified** the hypothesis.

Chapter takeaways. An A/B test uses random assignment to attribute a metric change to the change itself rather than to confounders or noise. Significance (“is it real?”) and effect size (“is it big enough to matter?”) are independent — decide on both. Compute the required sample size in advance from baseline, MDE, power, and significance; an underpowered test is wasted. Pre-commit the metric, sample size, duration, and decision rule, and read once — peeking and subgroup fishing manufacture false positives. Defend a read by naming what would have falsified the hypothesis.

5 Multi-Touch Attribution

5.1 The question every marketer cannot quite answer

A customer sees a Meta ad on Monday, reads a blog post on Wednesday, clicks a Google search ad on Friday, and buys on Sunday. Which channel “caused” the sale? The honest answer is **all of them and none of them alone** — but the budget must be split somehow, and how you split the credit determines where next quarter’s money goes. **Attribution** is the set of rules for assigning credit for a conversion across the touchpoints that preceded it. Every rule is a simplification, and every simplification has a **bias** that systematically distorts the allocation decision.

Definition 12 (Attribution model) An **attribution model** is a rule for distributing credit for a conversion across the marketing touchpoints in the customer’s journey. The chosen model determines the apparent return of each channel, and therefore the budget allocation — so the model is not a neutral measurement but an **assumption** baked into every spending decision.

5.2 Three models and their biases

Definition 13 (Last-click, linear, data-driven)

- **Last-click** assigns 100% of the credit to the final touch before conversion. Simple and verifiable, but **systematically over-credits** bottom-of-funnel channels (branded search, retargeting) and makes top-of-funnel awareness-builders look worthless.

- **Linear** (and other rule-based multi-touch: first-click, time-decay, position-based) splits credit across touches by a fixed rule — e.g. equally. It restores some credit to earlier touches but the split is **arbitrary**: there is no reason reality divides credit exactly evenly.
- **Data-driven** uses the data themselves (e.g. comparing converting vs non-converting paths, or experimental lift) to estimate each touch’s **incremental** contribution. The most principled, but **data-hungry**, **opaque**, and only as good as its inputs and assumptions.

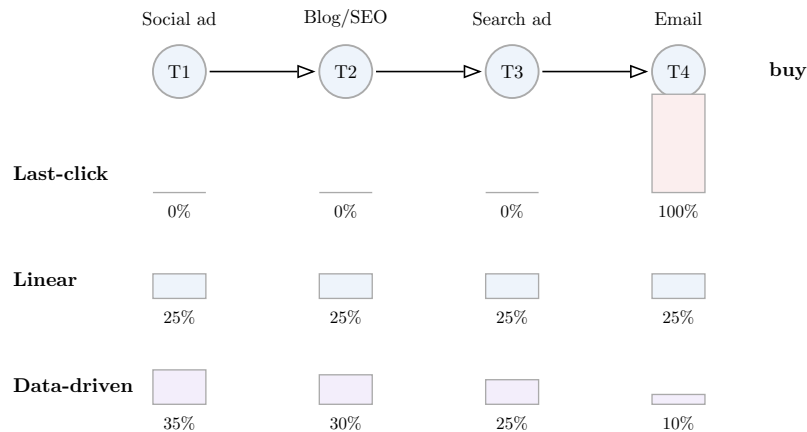


Figure 5: Three models split credit for the same four-touch journey very differently. Last-click hands everything to Email (the final touch); linear splits evenly by rule; data-driven estimates incremental contribution — here crediting the early Social ad and Blog most and the final Email least. The same data, three allocation decisions.

Worked example — How the model flips the decision. A brand spends equally on a top-of-funnel Social campaign and a bottom-of-funnel branded-Search campaign. Under **last-click**, branded search gets 90% of credit (it is usually the final touch), so the dashboard says “cut social, double search.” But branded search largely harvests demand that **social created** — under a **data-driven** model that measures incremental lift, social gets far more credit, and cutting it would quietly shrink the whole funnel. Same spend, same sales; opposite conclusions purely from the attribution rule.

5.3 Reasoning under attribution ambiguity

The mature stance is not to find the “true” model — there isn’t one, because credit assignment is partly unobservable — but to **choose a model suited to the context and reason explicitly about how its bias would change the decision**.

Common pitfall Last-click is the default in most analytics tools, and its bias is the most dangerous because it is invisible: it silently defunds every awareness and consideration channel, which over time starves the top of the funnel that bottom-of-funnel channels harvest from. If you must use last-click for simplicity, at least state the bias aloud and sanity-check big cuts against an incrementality test (Chapter 4) — the only way to measure true causal contribution is a controlled experiment, not any attribution rule.

Remark Attribution is where Chapters 4 and 5 meet. Attribution **models** divide observed credit using correlational data; an A/B / geo holdout test measures **incremental** causal lift directly. When the stakes are high, do not arbitrate between attribution models — run

an experiment that turns the channel off for a random subset and measures what actually changes.

Exercises

13. For the four-touch journey Social → Blog → Search → Email, write the credit split under last-click, first-click, linear, and a time-decay model. State which channel each model favours and the bias that produces that favouritism.

14. A company using last-click attribution concludes that branded search is its best channel and shifts budget into it from social. Explain the mechanism by which this decision could reduce total sales even though branded search “looks” most efficient.

15. Choose an attribution model for (a) a small DTC brand with a short purchase cycle and limited data, and (b) a large advertiser with long journeys and rich data. Justify each choice and name the bias you are accepting.

Chapter takeaways. An attribution model is a credit-assignment **assumption** that drives budget allocation. Last-click over-credits bottom-of-funnel channels and invisibly starves awareness; linear and other rule-based splits are less biased but arbitrary; data-driven estimates incremental contribution but is data-hungry and opaque. There is no “true” model — choose one suited to the context, state its bias, and reason about how that bias changes the decision. For high-stakes calls, measure causal lift with an experiment rather than trusting any attribution rule.

6 Customer Analytics: CLV, RFM, Cohorts, and Churn

6.1 From “how many customers” to “which customers, worth what”

Acquisition counts customers; this chapter values and segments them. Two businesses with identical customer counts can be worth wildly different amounts if one keeps customers for years and the other loses them in months. The tools here — lifetime value, RFM segmentation, cohort analysis, and churn signals — let you compute what a customer is worth, distinguish your best customers from your worst, watch how each **cohort** behaves over time, and catch customers who are about to leave while you can still act. All of them are, in practice, SQL queries (DAT22-5) over a transactions table.

6.2 Customer Lifetime Value

Definition 14 (Customer Lifetime Value (CLV / LTV)) CLV is the total contribution margin a customer generates over their entire relationship with the business. A simple, widely used form is

$$\text{CLV} = \frac{\text{ARPU} \times \text{gross margin}}{\text{churn rate}},$$

where ARPU is average revenue per user per period and the churn rate is the fraction of customers lost per period (so $\frac{1}{\text{churn}}$ is the expected lifetime in periods). More careful versions discount future cash flows to present value and use contribution margin rather than revenue.

Worked example — Computing CLV and closing the loop with CAC. A streaming service has ARPU of €10/month, 70% gross margin, and 5% monthly churn. Expected lifetime is $\frac{1}{0.05} = 20$ months, and

$$\text{CLV} = \frac{10 \times 0.70}{0.05} = \text{€}140.$$

Now recall the viability condition (Chapter 2): if blended CAC is €40, then $\frac{\text{CLV}}{\text{CAC}} = \frac{140}{40} = 3.5$ — healthy, with a payback period of $\frac{40}{10 \times 0.70} \approx 5.7$ months. Halving churn to 2.5% would **double** CLV to €280 — which is why retention work (Chapter 1) is often the highest-return lever in the whole business: it multiplies the value of every customer you already paid to acquire.

Common pitfall Two CLV errors. First, using **revenue** instead of **contribution margin** — CLV must net out the cost of serving the customer, or you will overpay for acquisition. Second, treating churn as constant when it varies by cohort and tenure (new customers churn faster than tenured ones); a single blended churn rate can badly misvalue a business whose early-life churn is severe. When in doubt, compute CLV per cohort (below) rather than one global number.

6.3 RFM segmentation

CLV gives one number per customer; RFM sorts customers into actionable groups using only their transaction history — exactly the kind of `GROUP BY` aggregation from DAT22-5.

Definition 15 (RFM segmentation) RFM scores each customer on three behavioural dimensions:

- **Recency** — how recently they last purchased (recent buyers are most likely to buy again);
- **Frequency** — how often they purchase;
- **Monetary** — how much they spend.

Each dimension is typically scored 1–5 (e.g. by quintile), giving segments like “champions” (5-5-5: recent, frequent, high-spend) and “at-risk” (1-5-5: once valuable, now gone quiet). The segments are **actionable**: each calls for a different marketing treatment.

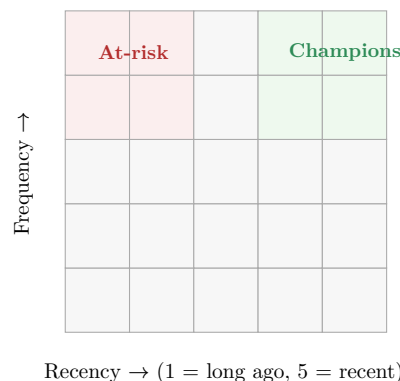


Figure 6: An RFM grid on the Recency–Frequency plane (Monetary as a third axis, omitted here). High-recency, high-frequency customers are **champions** (reward, upsell, ask for referrals); once-frequent customers who have gone quiet are **at-risk** (win-back campaigns). Each cell is a different action.

Worked example — Acting on RFM segments. A retailer scores its base. **Champions** (5-5-5) get early access and referral asks — not discounts they do not need. **At-risk** (low recency, high past frequency and spend) get a win-back offer **now**, before they are gone for good. **New high-spenders** (5-1-5) get onboarding to build the second purchase. **Hibernating low-value** (1-1-1) get little or no spend — chasing them rarely pays. Same budget, but directed by behaviour instead of sprayed evenly.

6.4 Cohort analysis

A single retention number hides whether the product is getting better or worse over time. Cohort analysis fixes this by grouping customers by **when they joined** and tracking each group separately.

Definition 16 (Cohort analysis) A **cohort** is a group of customers who share a start period (e.g. “signed up in March”). **Cohort analysis** tracks a metric — usually retention or revenue — for each cohort across the periods since it began, laid out as a **retention triangle**. Reading **down a column** compares cohorts at the same age (is the product improving for newer joiners?); reading **across a row** shows one cohort’s decay over time. It separates genuine product improvement from changes merely caused by mixing old and new customers.

	M0	M1	M2	M3
Jan	100%	60%	45%	38%
Feb	100%	65%	52%	
Mar	100%	72%		
Apr	100%	78%		

months since joining →

Figure 7: A retention triangle. Each row is a join-month cohort; each column is age in months. The triangle shape comes from newer cohorts having fewer elapsed months. Reading **down** the M1 column — 60%, 65%, 72%, 78% — shows month-1 retention improving cohort over cohort: the product is genuinely getting stickier, a signal a single blended retention number would have hidden.

Common pitfall Without cohorts, growth in total active users can **mask** worsening retention: a flood of new signups keeps the headline number up while each cohort actually decays faster than the last. Always cut retention by cohort before concluding the product is healthy. Conversely, a temporary dip in the blended number may just be one bad acquisition cohort, not a product regression — the triangle tells them apart.

6.5 Churn early-warning signals

Definition 17 (Churn and early-warning signals) **Churn** is a customer ceasing to be active (cancelling a subscription, or not purchasing within a defined window). **Early-**

warning signals are observable behaviours that **precede** churn and so allow intervention before it happens: declining usage frequency, longer gaps between purchases, falling order value, lapsed engagement (no logins/opens), or unresolved support tickets. The point is to act on the **leading** signal, not the **lagging** fact of churn — once a customer has left, win-back is far costlier than retention.

Worked example — A churn signal in the data. A SaaS tool notices that accounts whose weekly active-user count drops more than 30% for two consecutive weeks churn within 60 days about 70% of the time. That drop is an early-warning signal: a query flags such accounts, and customer success reaches out **while the contract is still live**. Acting on the lagging fact (the cancellation) would be too late; acting on the leading signal (the usage drop) saves accounts. This is RFM's **recency** logic generalised to engagement.

Remark Every tool in this chapter is a SQL query over a transactions or events table (DAT22-5): CLV is an aggregate with margin and churn; RFM is **GROUP BY** customer with recency, count, and sum; a cohort triangle is a **GROUP BY** on join-period and activity-period; a churn signal is a windowed comparison. Customer analytics is where your querying skill becomes marketing judgment.

Exercises

16. A subscription business has ARPU €25/month, 60% gross margin, and 4% monthly churn. Compute expected lifetime, CLV, and — given a €180 CAC — the LTV/CAC ratio and payback period. Then recompute CLV if churn is halved, and comment on which lever (lower CAC or lower churn) helps more here.

17. Given a small transactions table (customer, date, amount), describe the SQL you would write to assign each customer a Recency, Frequency, and Monetary quintile score, and name the marketing action for the 5-5-5 and 1-5-5 segments.

18. You are shown a retention triangle whose blended active-user count is rising but whose M1 retention falls cohort over cohort (60%, 55%, 50%, 45%). Explain what is really happening and why the headline number is misleading.

Chapter takeaways. CLV is lifetime contribution margin, $\approx \frac{\text{ARPU} \times \text{margin}}{\text{churn}}$; judged against CAC it tests viability and shows that cutting churn multiplies the value of every acquired customer. RFM segments customers by Recency, Frequency, Monetary into actionable groups (champions, at-risk). Cohort analysis groups customers by join period and tracks each over time, separating real product improvement from mix effects a blended number hides. Churn early-warning signals (declining usage, lengthening gaps) let you intervene before the customer leaves. All four are SQL queries over transaction data.

7 Performance vs Brand, and Marketing Automation

7.1 The tension that organises a marketing budget

Every marketing euro can be spent two ways. **Performance** marketing chases a measurable, near-term action — a click, a signup, a sale this week — and reports a clean CAC. **Brand** marketing builds durable mental availability and preference that pays off slowly and is hard to

attribute. The pull toward performance is strong precisely **because** it is measurable: it shows up in the dashboard tomorrow. But a business that spends only on performance slowly starves the demand that performance harvests — the same dynamic we saw in attribution (Chapter 5), now as a budget philosophy.

Definition 18 (Performance vs brand marketing) Performance marketing optimises for immediate, attributable conversions (search/social direct-response, retargeting); its strength is measurability and short payback, its weakness is that it mostly **captures existing demand**. Brand marketing builds long-term awareness, associations, and distinctiveness (Romaniuk/Sharp, BUS13-1); its strength is **creating future demand** and pricing power, its weakness is slow, hard-to-attribute payoff. The two are complements, not substitutes: brand fills the funnel that performance converts.

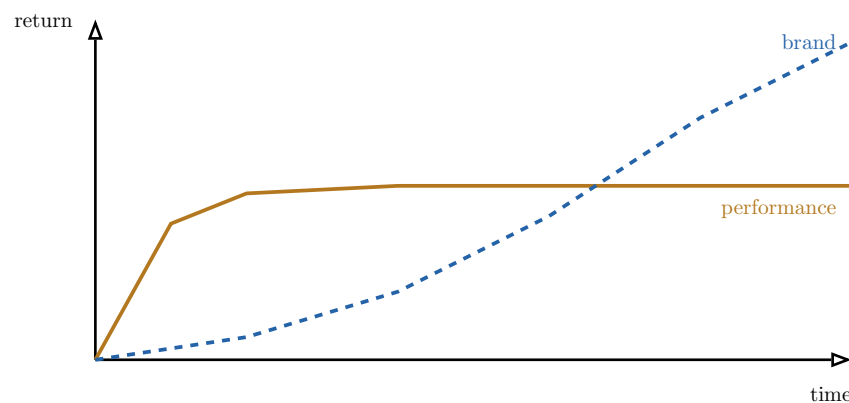


Figure 8: Performance marketing pays back fast then plateaus (it harvests existing demand); brand marketing pays back slowly but compounds (it creates new demand and pricing power). Optimising only the fast, measurable curve under-invests in the slow one — the long-term/short-term tension behind Binet & Field’s 60/40 brand-vs-activation benchmark from BUS13-1.

Common pitfall Because performance is measurable and brand is not, budgets drift toward performance until brand is quietly defunded — and then performance CAC slowly **rises** as the unbranded funnel thins. This is the marketing version of the last-click trap (Chapter 5): rewarding what is easy to measure and starving what is not. The corrective is the Binet & Field long/short discipline from BUS13-1 — commit a brand share of budget **as a rule**, not as the residual left after performance.

7.2 Marketing automation is execution, not strategy

The plumbing that **executes** segmented, lifecycle marketing — CRM, email/lifecycle flows, audience segmentation, triggered campaigns — is marketing automation. It is powerful and necessary, and it is also where teams most often mistake activity for strategy.

Definition 19 (Marketing automation) Marketing automation is the tooling (CRM, lifecycle email/SMS, segmentation engines, triggered campaigns) that executes marketing actions automatically based on customer data and behaviour — e.g. an onboarding sequence on signup, a win-back flow on a churn signal, an upsell on a usage milestone. It is an **execution layer** that scales decisions already made; it does not decide **which** segments matter, **what** the positioning is, or **whether** the unit economics work.

Common pitfall The recurring error is treating automation **as** strategy: buying a CRM and assuming “we now have a growth strategy.” Automation faithfully executes whatever logic you give it — including a bad segmentation, a wrong position, or an unprofitable channel, now at scale and on autopilot. Strategy (Where to Play / How to Win / How to Allocate, BUS13-1) decides **what** to automate; automation only decides **how fast**. A precise machine pointed at the wrong target just reaches the wrong place sooner.

Worked example — Automation amplifying a good vs a bad strategy. Two companies buy the same lifecycle-email platform. Company A has done the work: RFM segments (Chapter 6), a churn signal, a clear position. It wires automation to those — onboarding for new high-spenders, win-back on the churn signal, referral asks to champions — and results compound. Company B has no segmentation and an unclear position; it blasts the same newsletter to everyone on a schedule. Same tool, opposite outcomes: the automation amplified A’s strategy and amplified B’s **absence** of one.

Exercises

19. A company shifts its entire budget to performance marketing because brand spend “can’t be measured.” Predict what happens to performance CAC over the next year and explain the mechanism, connecting it to the attribution discussion of Chapter 5.
20. Design three automated lifecycle flows for a subscription product, each triggered by a specific customer-data event from Chapter 6 (e.g. activation, a churn early-warning signal, a champion threshold). State the trigger and the goal of each.
21. Explain, with an example, why buying a marketing-automation platform is not a marketing strategy, and what strategic decisions must be made **before** the tool can add value.

Chapter takeaways. Performance marketing is measurable and fast but mainly **captures** existing demand; brand marketing is slow and hard to attribute but **creates** future demand and pricing power — they are complements, and measurement bias drifts budgets toward performance until brand is starved and CAC rises (the Binet & Field 60/40 discipline from BUS13-1 is the corrective). Marketing automation (CRM, lifecycle flows, segmentation) is an **execution layer** that scales decisions already made; it amplifies a good strategy and a bad one alike, and is never a substitute for strategy.

8 Consumer Anthropology

8.1 Why people actually buy

So far we have measured **what** customers do. This chapter asks **why** — and the answer is rarely the obvious functional one. People do not buy a €5,000 watch to tell the time, or an electric drill because they want a drill. Understanding the real driver behind consumption lets you position, segment, and predict in ways that pure behavioural data cannot, and it is the layer that resurfaces in S6’s Strategic Due Diligence. We build it in three parts: **why** people consume (utility and social meaning), the **job** a product is hired to do (JTBD), and how genuinely new things **spread** (diffusion).

8.2 Beyond utility: distinction and identity

Definition 20 (Three layers of consumption value) Consumption serves more than function:

- **Utility** — the functional benefit (a coat keeps you warm). Classical economics stops here.
- **Social distinction** — consumption as a signal of status or group membership. **Veblen** goods are bought **because** they are expensive: the price **is** the point, signalling that one can afford waste (conspicuous consumption). **Bourdieu** showed taste itself is a marker of social class — what you consume classifies you.
- **Identity construction** — consumption as a way of building and expressing who one is (or wishes to be); brands become symbolic resources for the self.

Most purchases mix all three; premium and lifestyle categories are dominated by the second and third.

Worked example — The same category across the three layers. Consider bags. A €20 backpack is bought for **utility** — carry things. A €2,000 designer handbag of similar carrying capacity is bought for **distinction** — it signals status precisely because it is expensive (Veblen) and because recognising and valuing that brand marks a certain taste and milieu (Bourdieu). A sustainably-made canvas tote bought by someone who sees themselves as environmentally conscious is **identity construction** — it expresses a self. A marketer who prices and positions the handbag as if utility (capacity per euro) were the buying criterion has fundamentally misread the category.

Common pitfall The engineer's fallacy is to assume customers buy on **utility** and to compete on specs and price-per-feature. In status- and identity-driven categories this is a losing game: a **more functional, cheaper** product can lose decisively to one that better serves distinction or identity. Always ask which layer actually drives the category before competing on the wrong one.

8.3 Jobs-to-be-Done

Definition 21 (Jobs-to-be-Done (Christensen)) The **Jobs-to-be-Done** lens holds that customers “hire” a product to make progress on a **job** — a goal in a circumstance — rather than buying it for its attributes. The unit of analysis is the **job**, not the customer demographic or the product category. Competitors are then anything else that could be hired for the same job, including non-consumption and products from entirely different categories.

Worked example — The milkshake — the classic case. A fast-food chain tried to improve milkshake sales by making them tastier; nothing worked. Studying **when** people bought them revealed that a large share were bought in the morning by lone commuters — hiring the milkshake for the job “keep me occupied and not-hungry during a boring drive.” Its real competitors were bananas, bagels, and coffee, not other milkshakes. The improvements that mattered — thicker (lasts the whole commute), easy one-handed — followed only once the **job** was understood. The demographic (“men 18–35”) was useless; the **job** was everything.

Remark JTBD is the same demand-side, segment-by-the-job logic introduced in BUS13-1 — here sharpened into the durable analytical core that later resurfaces as commercial-due-diligence logic. A market defined by **jobs** is far more stable than one defined by current products: products come and go, but the underlying job (“get me through the morning commute”) persists.

8.4 Diffusion of innovations: trend vs fad

New products do not reach everyone at once; they spread through a population in a characteristic pattern, and reading that pattern is how you tell a **structural trend** from a **fad**.

Definition 22 (Diffusion of innovations (Rogers)) Innovations spread through five adopter groups in sequence — **innovators** (2.5%), **early adopters** (13.5%), **early majority** (34%), **late majority** (34%), and **laggards** (16%) — producing an S-shaped cumulative adoption curve. The decisive test of a durable trend is whether an innovation **crosses** from early adopters to the early majority; many fail at this gap and fade. A **trend** clears the gap and reaches the majority; a **fad** spikes among early groups and never crosses.

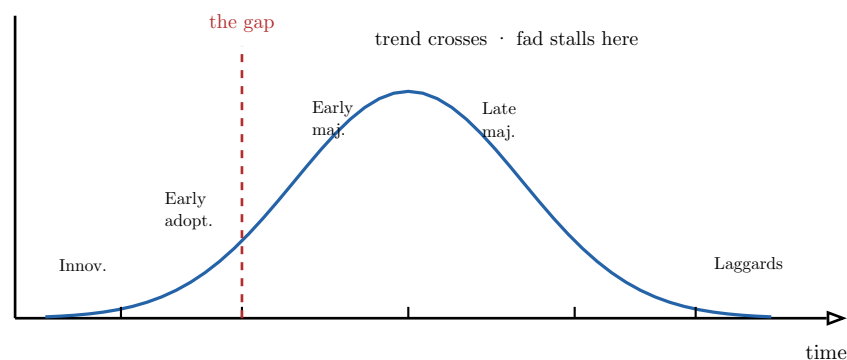


Figure 9: Rogers’ adopter distribution. The decisive question for trend-vs-fad is whether the innovation crosses the gap (dashed line) from early adopters to the early majority. A **structural trend** crosses and reaches the majority; a **fad** peaks among innovators and early adopters and never makes the jump.

Common pitfall Do not mistake **early-adopter enthusiasm** for a **proven trend**. Innovators and early adopters try almost anything new — their excitement is a weak signal, because it precedes the gap that kills most fads. The real evidence of a structural trend is **crossing into the early majority**: adoption by pragmatic buyers who need a reason beyond novelty. Conversely, do not dismiss something as a fad merely because it is new; use the diffusion test and the JTBD question (does it serve a durable job?) rather than instinct.

Exercises

22. Choose a product or category and analyse it across the three consumption layers (utility, distinction, identity), saying which dominates and what that implies for how it should be priced and positioned.

23. Pick a product and state the **job** customers hire it to do, using a JTBD formulation. List two competitors from **different categories** that are hired for the same job, and one form of non-consumption it competes against.

24. Take a current phenomenon you suspect is a fad and one you believe is a structural trend. For each, apply the diffusion test (has it crossed into the early majority?) and the JTBD test (does it serve a durable job?) to defend your classification with evidence rather than instinct.

Chapter takeaways. Consumption serves utility, social distinction (Veblen’s conspicuous consumption, Bourdieu’s taste-as-class), and identity construction — and premium/lifestyle categories are driven by the latter two, so competing on utility alone misreads them. Jobs-to-be-Done (Christensen) analyses the **job** a product is hired for, not the demographic or category, which reveals true competitors and durable markets — the seed of S6 due-diligence logic. Rogers’ diffusion spreads innovations through five adopter groups; the test of a real trend versus a fad is whether it **crosses** from early adopters into the early majority.

9 The Integrated Growth Plan

9.1 Putting the whole course to work

The final task is the one every prior chapter was building toward: a complete, data-grounded growth plan for a real company. It is not an essay of tactics but a defended set of **allocation decisions** — which channels, how much budget, measured by what dashboard, projected over twelve months, and stress-tested against the assumptions that could be wrong. This chapter assembles the pieces.

Definition 23 (The integrated growth plan) An **integrated growth plan** for a company specifies, with evidence:

1. **Diagnosis** — the AARRR funnel and its weakest stage, and the growth loop(s) with their constraints (Chapters 1, 3).
2. **Channel selection** — a portfolio of acquisition channels chosen on LTV/CAC economics and audience fit, with saturation points (Chapter 2).
3. **Budget allocation** — how the marketing budget is split across channels and across brand vs performance (Chapters 2, 7), with the marginal logic for each euro.
4. **KPI dashboard** — the few metrics that govern the plan: CAC by channel, LTV, LTV/CAC, payback, activation and retention rates, cohort curves (Chapters 1, 2, 6).
5. **12-month CAC/LTV projection** — a forward model of customers, CAC, and contribution, with **explicit assumptions** and a **sensitivity analysis** across assumption ranges (this chapter).

9.2 Channel selection and budget allocation

Channel selection is the portfolio decision from Chapter 2 made concrete: rank channels by marginal LTV/CAC and audience fit, fund each up to its saturation point, and split the residual between brand and performance per the Binet & Field discipline (Chapter 7). Allocation is marginal, not average: the next euro goes wherever its marginal customer is cheapest relative to LTV, until that channel saturates and the next channel becomes the best marginal buy.

Worked example — A marginal allocation. A DTC brand has €100k/month. Paid search returns customers at €25 CAC but saturates at €30k spend (≈1,000 customers); beyond that, marginal CAC rises past €60. Paid social returns €40 CAC up to €40k spend. SEO/content is

€10 CAC but capped by content velocity at 300 customers/month. The plan funds SEO to its cap first (cheapest), search to its saturation, social next, and holds 20% for brand — **not** an even split across channels, but a marginal one that respects each channel’s saturation curve.

9.3 The KPI dashboard

Definition 24 (Growth KPI dashboard) A **KPI dashboard** is the small set of governing metrics, reported per channel and per cohort, that tells whether the plan is working: **CAC** (paid, by channel), **LTV** and **LTV/CAC**, **payback period**, **activation** and **retention** rates, cohort retention curves, and the growth-loop coefficient. The discipline is to track the **few** metrics that drive allocation decisions, not a wall of vanity metrics (total signups, impressions) that move without changing any decision.

Common pitfall A dashboard full of **vanity metrics** — cumulative downloads, page views, social followers — feels informative and changes nothing, because none of them maps to an allocation decision. Keep a metric only if a change in it would change what you **do**. CAC by channel, LTV/CAC, and cohort retention earn their place; “total registered users since launch” does not.

9.4 The 12-month projection and sensitivity analysis

The plan’s quantitative core is a forward projection — and its credibility lives entirely in **making the assumptions explicit and testing how the conclusion moves when they are wrong**.

Definition 25 (Projection and sensitivity analysis) A **12-month CAC/LTV projection** models, month by month, customers acquired per channel (with rising CAC as channels saturate and any loop amplification), retention/churn, and the resulting contribution margin, against the budget. A **sensitivity analysis** re-runs the projection across plausible **ranges** of the key assumptions — CAC, churn, conversion, loop coefficient — to show which assumptions the conclusion actually depends on, and over what range the plan remains viable. It converts a single fragile forecast into a defended decision.

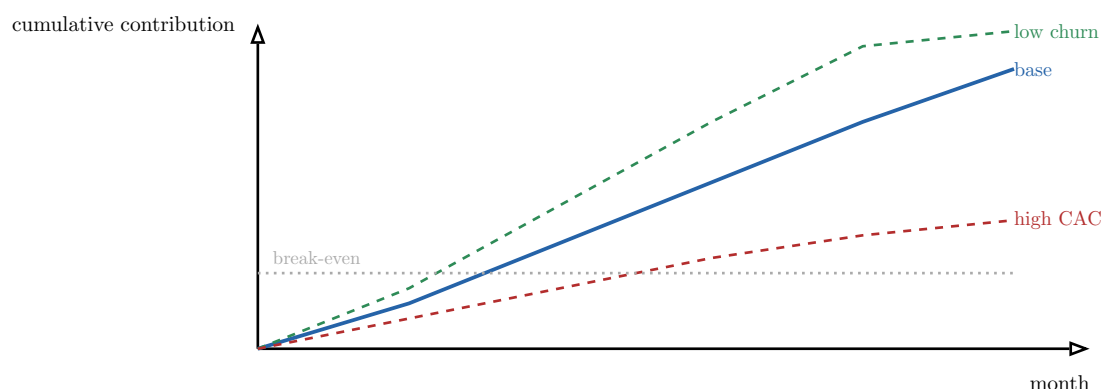


Figure 10: A 12-month projection with sensitivity bands. The base case (solid) and two stressed scenarios — lower churn (optimistic) and higher CAC from faster saturation (pessimistic) — show the range of outcomes. Because even the pessimistic case clears break-even, the plan is **robust** to its riskiest assumption; if it did not, the plan would depend on a number you cannot guarantee, and should be defended differently.

Worked example — Defending the plan against a constraint. The plan projects 12-month contribution of €450k on €600k spend, base case. The riskiest assumption is paid-search CAC staying at €25 — but it saturates at €30k spend. The sensitivity analysis shows that if search saturates 30% earlier than assumed, blended CAC rises and contribution falls to €310k — still profitable, but the plan now leans harder on SEO and the referral loop. The defensible recommendation is therefore: fund search only to its saturation point, invest early in the (slower) SEO and loop so they are producing before search tops out, and treat the loop constraint (network exhaustion, Chapter 3) as the thing to watch. The recommendation is quantified, and defended against the observable business constraint that could break it.

Remark This plan is the whole course in one artefact. AARRR locates the leak; channel economics and the LTV/CAC condition choose where to spend; loops decide what compounds; experimentation proves the fixes; attribution and the brand/performance split govern measurement honesty; customer analytics supplies LTV, churn, and cohorts; consumer anthropology tells durable jobs and trends from fads. Marketing has become what the preface promised: allocation decisions defended with evidence.

Exercises

25. For a real company of your choice, draft the channel-selection and budget-allocation sections of a growth plan: rank at least three channels by marginal LTV/CAC and audience fit, allocate a stated monthly budget across them and brand/performance, and justify each allocation marginally (not evenly).

26. Design a one-screen KPI dashboard for that company: list the 6–8 metrics you would track and, for each, name the allocation decision it informs. Then list two vanity metrics you deliberately exclude and why.

27. Build a simple 12-month projection (it can be a table) for the plan, stating every key assumption (CAC by channel and its saturation, churn, conversion, loop coefficient). Run a sensitivity analysis on the two assumptions you judge riskiest and state over what range the plan remains viable — and what you would change if the pessimistic case broke break-even.

Chapter takeaways. An integrated growth plan is a defended set of allocation decisions: diagnose with AARRR and loops; select a **portfolio** of channels on marginal LTV/CAC and audience fit up to saturation; allocate budget marginally and split brand vs performance deliberately; govern with a small KPI dashboard of decision-driving metrics (not vanity counts); and project 12 months of CAC, LTV, and contribution with **explicit assumptions** and a **sensitivity analysis**. The plan's credibility comes from quantified recommendations defended against the observable constraints — saturation, churn, loop limits — that could break them.